

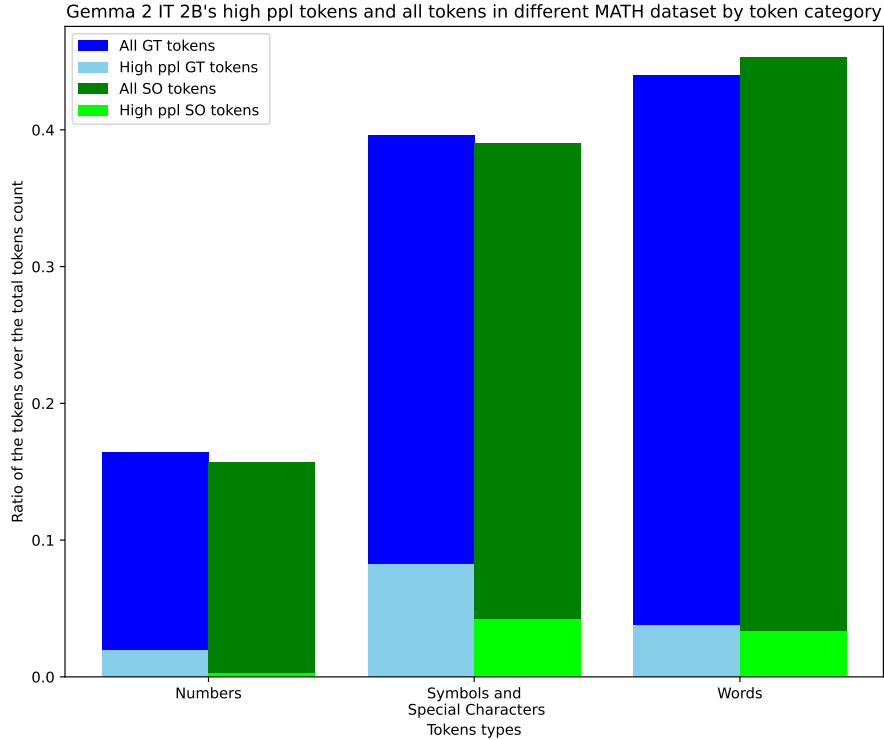


Figure 10: Ratio changes of high perplexity tokens before and after STM filtering on ground-truth (GT) dataset and Self-Output (SO) dataset of MATH task with Gemma 2 IT 2B

In BIRD task, we basically follow the same prompt as [44] does. A database schema (schema) is followed by a text-to-sql instruction and the question query (question) as the Textbox below shows:

BIRD prompt

user prompt {schema}
 -- Using valid SQLite, answer the following question for the tables provided above.
 -- Question: {question}
 Now, generate the correct SQL code directly (Do NOT generate other text except the SQL code):

G Other STM alternatives

Besides cross-scale setting, inspired by [20, 25], we also investigate possible STM settings that acquire a model trained on ground truth data:

Differential Perplexity Filtering (DPF). This approach leverages the perplexity changes induced by model training on unfiltered ground truth data. By computing the perplexity differential between the base and fine-tuned models, we identify tokens that demonstrate improved learnability during unrestricted training. The final training set is then constructed with tokens that exhibited reduced perplexity scores after fine-tuning, which potentially captures naturally learnable patterns in the data.

STM with DPF. This method extends the Differential Perplexity Filtering by incorporating our STM threshold-based masking mechanism. By combining both approaches, we aim to benefit from both the

Table 18: List of alternative token selection methodology supervised fine-tuned on MATH in Gemma 2 IT 2B . Here we choose Gemma 2 series because Gemma 2 has feasible model sizes to implement the cross-scale setting for comparison.

GEMMA 2 IT 2B	FILTER	MATH	MBPP	GSM8K	ARC	BIRD
ORIGINAL MODEL	0%	28.1	51.1	58.5	74.7	13.9
BASLINE FINE-TUNING	0%	21.7	45.8	19.0	29.5	5.1
<i>STM</i> $\tau = 2.5$	23.8%	30.3	49.5	55.4	75.3	13.3
<i>STM</i> _{9B} $\tau = 2.0$	21.4%	27.2	47.6	42.0	72.4	14.1
<i>DPF</i> $\tau = 0$	39.2%	22.5	40.2	1.1	6.0	11.0
<i>STM + DPF</i> $\tau = 100$	44.7%	17.3	43.7	9.1	14.4	11.7

Table 19: STM applied on continual learning settings

STEPS	TRAINING TASK	MATH-TEST	MBPP-TEST	GSM8K	ARC	BIRD
0	INITIAL MODEL	0.3140	0.5979	0.7255	0.7816	0.2053
1	MATH-TRAIN	0.334	0.6164	0.7687	0.7577	0.2027
2	MBPP-TRAIN	0.332	0.625	0.7635	0.7688	0.2073

identification of learnable tokens through SFT and the prevention of high-perplexity token influence through STM, potentially offering a more robust token selection strategy than either method alone.

Table 18 shows the results for the best-performance models of the original STM and the other alternative of STM. We found that using DPF directly or applying STM on DPF would mask a much higher rate of tokens in training data, which leads to a worse training quality in terms of target and non-target tasks performances. One setting different from the previous works is that we did not prepare a high quality or high scored data for the reference model to train. Instead, we use ground truth data to filter the "unlearnable" tokens out. However, even though we mask out unlearnable tokens out, the remaining unfiltered tokens are still too few to maintain good quality of training.

H A continual learning setting of STM

We have experimented the effectiveness of STM by training on a single target task to reduce non-target tasks degradation. Yet, in a setting of continual learning scenario as [14] presents, degradation of tasks also occurs after several target tasks are trained sequentially. We simulate such setting to fine-tune Llama 3 8B Instruct with MATH initially and followed by MBPP tasks. Table 19 shows that (1) not only the target task is improved after fine-tuning, but also its degradation is very small after training another target task; (2) the non-target tasks like GSM8K, ARC and BIRD keeps similar performance even after two rounds of fine-tuning. It indicates STM can be effective in such continual learning setting as well.

I Does self-output response correctness matters?

In our experiment, the lowest perplexity scored data are collected exclusively from correct Self-Output subsets. In MBPP training task, a correct Self-Output data refers to a generated response that passes all the test cases. In MATH, correctness refers to a correct final answer parsed from a response. Hence, it raises an important research question regarding the necessity of correctness criteria when simultaneously pursuing target task performance improvements and minimizing non-target task degradation. In Table 20, we collect different correctness rates of Self-Output response as MBPP training datasets respectively with Llama 3 8B Instruct model. Correctness rate refers to the ratio of the samples whose labels are verified as "correct". Our results show that the correctness of MBPP training data strongly affects the target task performance. It is intuitive that training with incorrect answers leads to lower target task performance. However, compared to model trained on ground

Table 20: The performance of models trained with different correctness rate of MBPP Self-Output dataset and trainable tokens number. Values of a cell block in **gray** represents the target task testing performance of the Llama 3 8B Instruct trained with target task data.

REJECTION RATE	TRAINED TOKENS	MBPP	MATH	ARC	GSM8k	BIRD
BASELINE FINE-TUNING	75,464	.579	.233	.591	.185	.176
SELF-OUTPUT						
100%	65,602	.606	.324	.793	.773	.199
75%	69,605	.593	.346	.794	.778	.199
0%	59,738	.561	.315	.802	.770	.188
ORIGINAL MODEL	-	.593	.314	.782	.726	.206

Table 21: The performances of models trained with different correctness rate of MBPP Self-Output dataset and trainable tokens number. Values of a cell block in **gray** represents the target task testing performance of the Gemma 2 IT 2B trained with target task data.

REJECTION RATE	TRAINED TOKENS	MBPP	MATH	ARC	GSM8k	BIRD
BASELINE FINE-TUNING	86,632	.400	.193	.229	.716	.051
SELF-OUTPUT						
100%	87,632	.500	.287	.742	.563	.134
49%	90,594	.489	.286	.732	.564	.134
0%	90,416	.497	.284	.742	.563	.135
ORIGINAL MODEL	-	.511	.281	.747	.585	.139

truth data (the correctness rate is 100%), which fails to sustain non-target task performance, models trained with higher correctness rate of Self-Output data achieves higher target task improvements and still sustains the non-target task performance robustness at the same time. Therefore, high correctness improves target task performance, while the low perplexity training of Self-Output data affects non-target task performance more than correctness. We have also conducted the correctness test with smaller models like Gemma 2 IT 2B Instruct, please refer to Appendix I.

We have also tested the ablation test of correctness with Gemma 2 IT 2B model as Table 21 shows. It holds the same conclusion that correctness is relevant to target task performance only, while the low perplexity trait of Self-Output data is more relevant to non-target task performance than correctness. However in Gemma 2 IT 2B setting, the number of training instance with at least one positive and one negative responses at the same time is much fewer than ground truth data and Llama 3 8B Instruct’s correctness data although it has more trainable tokens due to different tokenizers are applied. That is to say, to maintain the same training data amount between each correctness rate dataset for comparison in the Figure, we have to choose a smaller training data size for Gemma 2 IT 2B. Therefore, the effectiveness of data correctness is not as obvious as Llama 3 8B Instruct model does.

J Regularization effects on non-target tasks

We apply a range for searching best combination of weight decay and dropout rate hyperparameters when training an MBPP task on Llama 3 8B Instruct model. The weight decay is set from 0.0 to 0.5 with interval 0.1, and dropout rate is set from 0.05 to 0.5 with an interval of 0.05 to 0.1. And we pick the top 3 best performing weight decay value with its dropout rate, and choose the best 2 performances among the checkpoints of 50 steps to 300 steps (about 5 to 6 epochs training) as our results. Our experiment results as Figure 11 show that, when applying dropout or weight decay on ground truth data, the model trained with target task like MBPP still overfits on training data and the testing performance degrades. On the other hand, the non-target performance of MATH as Figure 12